# Contest Problem Set 12224

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# Number Sense 12224

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Name _____

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1. $222 + 21 =$ _____.
2. $631 - 469 =$ _____.
3. The units digit of 1312021 is _____.
4. $12 + 12 \times 4 =$ _____.
5. $32 + 45 + 55 + 38 =$ _____.
6. The remainder of $279 \div 5$ is _____.
7. $16 \times 30 =$ _____.
8. $429 + 285 =$ _____.
9. $156 \div 13 =$ _____.
10. (estimate) $492 + 1066 + 902 =$ _____.
11. $62 \times 11 =$ _____.
12. $561 - 138 - 218 =$ _____.
13. The remainder of $352 \div 9$ is _____.
14. $18 \times 5 \times 11 =$ _____.
15. $19^2 =$ _____.
16. $33 \times 27 =$ _____.
17. $36 \times 45 =$ _____.
18. *CCCXLVII* in Arabic numerals is _____.
19. $19.2 \div 1.2 =$ _____.
20. (estimate) $149 \times 601 =$ _____.
21. $25 \times 46 =$ _____.
22. The lesser of $\frac{6}{11}$ and $\frac{5}{9}$ is _____ (fraction).
23. $15 \times 49 \div 35 =$ _____.
24. $31^2 =$ _____.
25. The GCD of 35 and 55 is _____.
26. $544 \div 32 =$ _____.
27. $25 + 23 + 21 + 19 + 17 =$ _____.
28. $\frac{5}{6} - \frac{1}{12} - \frac{1}{4} =$ _____ (fraction).
29. If 1 yard is equal to 3 feet, then 5280 feet is equal to _____ yards.
30. (estimate) $89 \times 11 \times 91 =$ _____.
31. $99 \times 51 =$ _____.
32. $55^2 =$ _____.
33. $\frac{7}{10} =$ _____ %.
34. The perimeter of a square with area 196 is ___.
35. $39 \times 31 =$ _____.
36. The number of odd whole numbers between 92 and 184 is _____.
37. 25% of 76 is _____.
38. The LCM of 35 and 55 is _____.
39. $\sqrt{841} =$ _____.
40. (estimate) $\sqrt{262396} =$ _____.

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41. The remainder of $3755 \div 11$ is _____.

42. $88 \times 28 =$ _____.

43. $36^2 - 25^2 =$ _____.

44. The twelfth term in the arithmetic sequence $13, 28, 43, \ldots$ is _____.

45. $8 \times 14 + 12 \times 4 =$ _____.

46. The diameter of a circle with area $49\pi$ is _____.

47. $75 \times 101 =$ _____.

48. $144_8$ in base 10 is _____.

49. $2^5 \cdot 5^2 =$ _____.

50. (estimate) $27 \times 4444 =$ _____.

51. $103 \times 129 =$ _____.

52. The measure of an interior angle in a regular octagon is _____ $^\circ$.

53. $8\frac{1}{3}\%$ of 132 is _____.

54. The sum of the terms of the arithmetic sequence $7, 10, 13, \ldots 64$ is _____.

55. $12 \times 333 =$ _____.

56. The mean of the list $15, 21, 9, 34, 16$ is _____.

57. $10110_2$ in base 8 is _____ $_8$.

58. The area of a right triangle with a leg of length 5 and a hypotenuse of length 13 is _____.

59. $96 \times 82 =$ _____.

60. (estimate) $7338768 \div 269 =$ _____.

61. The number of ways to choose 2 toys from a set of 6 different toys is _____.

62. $0.1\overline{3} =$ _____ (fraction).

63. $5\frac{1}{6} \times 7\frac{1}{6} =$ _____ (mixed number).

64. The number 93 in base 5 is _____ $_5$.

65. $\sqrt[3]{140608} =$ _____.

66. $99 \times 125 =$ _____.

67. If $w = 11$, then $w^2 + 8w + 16 =$ _____.

68. $16 \times 17 \times 42 =$ _____.

69. The area of a rhombus with diagonals of length 50 and 64 is _____.

70. (estimate) $31^4 =$ _____.

71. $\sqrt{20} \times \sqrt{90} \times \sqrt{2} =$ _____.

72. $185^2 =$ _____.

73. If $5^{1-x} = \frac{1}{5}$, then $5^{1+x} =$ _____.

74. The volume of a cube with a surface area of 864 is _____.

75. $13^3 =$ _____.

76. The number of positive whole number divisors of 98 is _____.

77. $729^{\frac{2}{3}} =$ _____.

78. $34 \times 15 \times 49 =$ _____.

79. The sum of the terms of the infinite geometric sequence $36, 24, 16, \ldots$ is _____.

80. (estimate) $108^4 \div 9^2 =$ _____.

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# Sprint Round 12224

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. | A | B | C | D | E | 11. | A | B | C | D | E | 21. | A | B | C | D | E |
| 2. | A | B | C | D | E | 12. | A | B | C | D | E | 22. | A | B | C | D | E |
| 3. | A | B | C | D | E | 13. | A | B | C | D | E | 23. | A | B | C | D | E |
| 4. | A | B | C | D | E | 14. | A | B | C | D | E | 24. | A | B | C | D | E |
| 5. | A | B | C | D | E | 15. | A | B | C | D | E | 25. | A | B | C | D | E |
| 6. | A | B | C | D | E | 16. | A | B | C | D | E | 26. | A | B | C | D | E |
| 7. | A | B | C | D | E | 17. | A | B | C | D | E | 27. | A | B | C | D | E |
| 8. | A | B | C | D | E | 18. | A | B | C | D | E | 28. | A | B | C | D | E |
| 9. | A | B | C | D | E | 19. | A | B | C | D | E | 29. | A | B | C | D | E |
| 10. | A | B | C | D | E | 20. | A | B | C | D | E | 30. | A | B | C | D | E |

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1. How many of the products $4 \times 8$, $5 \times 5$, $11 \times 3$, $2 \times 15$, and $7 \times 4$ are greater than 31?

(A) 5 (B) 3 (C) 2 (D) 4 (E) 1

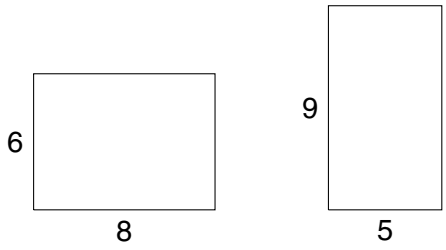
2. Hayden was given \$20 by his mother to purchase snacks this week. On Monday, Hayden spent \$4 on snacks. On Tuesday and Wednesday, Hayden spent \$3 each day on snacks. On Thursday, Hayden spent \$4 on snacks, and on Friday Hayden spent \$5 on snacks. How many dollars of the \$20 remained after Hayden bought snacks on Friday?

(A) \$1 (B) \$4 (C) \$2 (D) \$0 (E) \$3

3. What is the value of $11 + 15 + 19 + 21 + 25 + 29 + 31 + 35 + 39 + 41 + 45 + 49$?

(A) 300 (B) 380 (C) 400 (D) 360 (E) 320

4. One rectangle shown below has a width of 8 and height of 6. Another rectangle has a width of 5 and a height of 9. What is the total area of both rectangles?




(A) 87 (B) 95 (C) 98 (D) 93 (E) 88

5. Gary went to a showing of the latest *Ghostblasters* movie. The movie began at 7:18 PM and lasted for 124 minutes. At what time did the movie end?

(A) 9:22 PM (B) 9:24 PM (C) 9:26 PM (D) 9:18 PM (E) 9:28 PM

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6. Michelle is selling lemonade. She has five gallons of lemonade, and is selling the lemonade at a price of $2 per pint. If there are two pints in one quart, and four quarts in one gallon, then how many dollars will Michelle receive for selling all of her lemonade?

(A) $80 (B) $160 (C) $40 (D) $20 (E) $120

7. January 1st, 2022 was a Saturday. On what day of January 2022 does the last Wednesday of January 2022 occur?

(A) 27 (B) 28 (C) 29 (D) 25 (E) 26

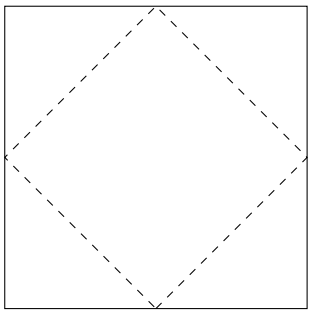
8. A prime number is a whole number that is both greater than 1 and only evenly disible by 1 and itself. For example, since 5 is only divisible by 1 and 5, the number 5 is a prime number. Of the answer choices, which number is not prime?

(A) 73 (B) 97 (C) 83 (D) 91 (E) 79

9. The product of two positive whole numbers is 84. When one number is subtracted from the other, the difference is 8. What is the sum of the two numbers?

(A) 24 (B) 20 (C) 22 (D) 18 (E) 26

10. Cindy takes a square piece of paper measuring 12 inches on a side and makes four cuts, with each cut beginning and ending at the middle point of one of the sides of the paper. After she is done, she then has four triangles and a smaller square. What is the area of the smaller square, in square inches?



12

(A) 96 (B) 60 (C) 90 (D) 84 (E) 72

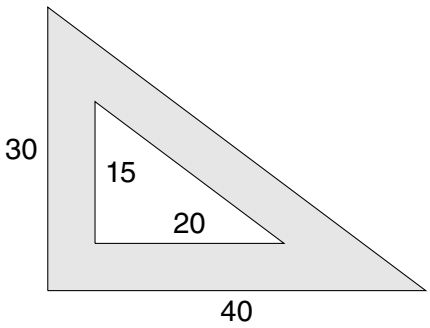
11. How many positive whole numbers evenly divide 165 but do not evenly divide 105?

(A) 3 (B) 4 (C) 5 (D) 6 (E) 2

12. Colleen is collecting acorns. She collects 3 acorns on the first day, 7 acorns on the second day, 11 acorns on the third day, and so on, collecting four more acorns each day than she did on the previous day. On what day of her collecting will her total collection have more than 299 acorns?

(A) 11 (B) 14 (C) 15 (D) 13 (E) 12

13. A piece of paper is in the shape of a right triangle with legs of length 40 centimeters and 30 centimeters. From the middle portion of this piece of paper, a piece of paper in the shape of a right triangle with legs of length 20 centimeters and 15 centimeters is cut out and removed. What is the area of the remaining piece of paper, in square centimeters?




(A) 450 (B) 480 (C) 550 (D) 400 (E) 420

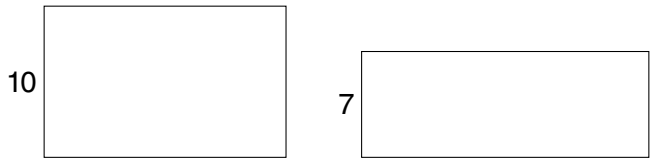
14. The Ryde ride service charges users a fee of $5 for any ride, and this fee also pays for the first mile of travel. After the first mile, users are charged a fee of $1.25 per mile traveled. Andy is using Ryde to travel to a football stadium that is seventeen miles from his home. In addition to the fees for his Ryde ride, Andy plans to pay his Ryde driver a $4 tip. What is the total cost, in dollars, of Andy's ride to the football stadium using Ryde?

(A) $31.50 (B) $25.25 (C) $28.00 (D) $30.25 (E) $29.00

15. The factorial of a positive whole number is written as a ! following the number, and it is calculated by multiplying all of the positive whole numbers from that number down to 1, inclusive. For example, 4! is equal to $4 \times 3 \times 2 \times 1 = 24$. What is the value of $(9! + 8! + 7!) \div (7!)$?

(A) 99 (B) 80 (C) 100 (D) 81 (E) 90

16. Two rectangles have the same perimeter. One rectangle has a height of 10, and the other rectangle has a height of 7. The area of the larger rectangle exceeds the area of the smaller rectangle by 27. What is the perimeter of one of the rectangles?




(A) 58 (B) 52 (C) 54 (D) 56 (E) 50

17. In the $3 \times 3$ grid of squares below, each of the whole numbers from 1 through 9, inclusive, is placed into one of the nine squares of the grid so that the sum of the numbers in any row or column of the grid is odd. Five of the squares already have numbers. In how many different ways can the remaining four numbers be placed into the remaining four squares?

| | | |
|---|---|---|
| 2 | 6 | 3 |
| 1 | | |
| 4 | | |

(A) 8 (B) 9 (C) 12 (D) 6 (E) 24

18. What whole number is nearest in value to $17\sqrt{7}$?

(A) 45 (B) 44 (C) 48 (D) 46 (E) 47

19. Stan purchased a share of stock at the end of the day on Monday. On Tuesday, the value of the stock increased by 20%, and on Wednesday, the value of the stock increased by 25%. But on Thursday, the value of the stock only increased by 10%, and finally on Friday the stock lost 40% of its value. Stan incorrectly thought that since $20\% + 25\% + 10\% - 40\% = 15\%$, the stock increased by 15% since he bought it. But the stock actually lost value since he made his purchase. By what percent had the value of the stock decreased since Stan made his purchase?

(A) 2% (B) 10% (C) 5% (D) 1% (E) 4%

20. If $a \clubsuit b = a^3 + 3a^2b + 3ab^2 + b^3$, then what is the value of $9 \clubsuit 5$?

(A) 2704 (B) 2854 (C) 2744 (D) 2684 (E) 2844

21. How many different arrangements of the letters of the word *SNOWS* either begin with an *S* or end with an *S* (possibly both)?

(A) 54 (B) 42 (C) 48 (D) 60 (E) 36

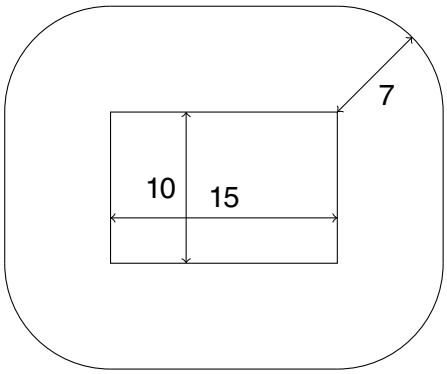
22. Henry and Isaac are riding their bikes on the same route. Henry rides his bike at a constant speed of 10 miles per hour, and Isaac rides his bike at a constant speed of 12 miles per hour. At noon, Henry rides his bike past a road sign. Exactly twelve minutes later, Isaac rides his bike past the same sign. At what time will Isaac catch up to Henry?

(A) 1:06 PM (B) 1:30 PM (C) 1:20PM (D) 1:12 PM (E) 1:15 PM

23. What is the remainder when $35 \times 36 \times 37 \times 38 \times 39$ is divided by 25?

(A) 20 (B) 10 (C) 5 (D) 0 (E) 15

24. A rectangular picture that has a width of 15 centimeters and a height of 10 centimeters is surrounded by a picture frame. Every point on the outer edge of the frame is 7 centimeters from the nearest point on the outer edge of the picture. To the nearest whole number, what is the area of the frame, in square centimeters?




(A) 496 (B) 512 (C) 508 (D) 504 (E) 511

25. Stella has a magic bag that contains an endless amount of red, white, and blue beads to use to make bracelets. She removes beads from the bag one at a time, and when she has three beads of any one color, she sets that group of three beads aside for use on a bracelet. What is the fewest number of beads that Stella must remove from the bag in order to guarantee that she has at least six groups of beads for use on bracelets?

(A) 18 (B) 19 (C) 22 (D) 20 (E) 21

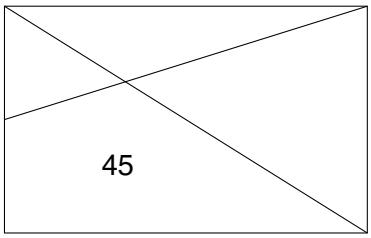
26. What is the smallest power of 5 that is greater than $2^{30}$?

(A) $5^{17}$ (B) $5^{15}$ (C) $5^{14}$ (D) $5^{13}$ (E) $5^{16}$

27. Micah and Noah are playing a game. A bag has 100 jellybeans. Micah and Noah take turns removing either 1, 2, 3, 4, 5, or 6 jellybeans from the bag. The player that removes the last jellybean loses. Micah gets to draw from the bag first, and realizes that he has a plan that guarantees that he wins the game by drawing a certain number of jellybeans on his first turn. How many jellybeans should Micah draw on his first turn to guarantee that he wins the game?

(A) 3 (B) 4 (C) 1 (D) 5 (E) 2

28. In the rectangle below, a diagonal intersects a segment that connects one vertex of the rectangle to the midpoint of a side. The diagonal and segment divide the rectangle into three triangles and a quadrilateral. The area of the quadrilateral is 45. The sum of the areas of the three triangles is a positive whole number. What is the sum of the digits of that positive whole number?




(A) 9 (B) 17 (C) 15 (D) 12 (E) 10

29. Kim chose a four-digit positive whole number. She noticed that the sum of the digits of the number was equal to the product of the digits of the number. How many different four-digit positive whole numbers could Kim have chosen?

(A) 24 (B) 16 (C) 8 (D) 6 (E) 12

30. The octal, or base 8 repeating value $0.1\overline{52}_8$ is equivalent to a fraction whose numerator and denominator are in base 10. What is the sum of the numerator and denominator of that fraction?

(A) 24 (B) 25 (C) 29 (D) 17 (E) 19

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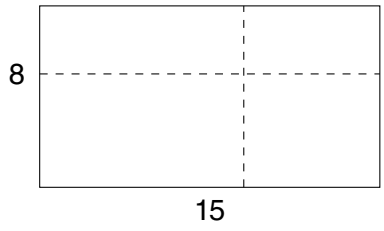
Problems 1 & 2

1. Amber's history test had twenty multiple-choice questions. Each multiple-choice question was worth three points. Her history test also had eight short-answer questions. Each short-answer question was worth five points. Amber answered seventeen of the multiple-choice questions correctly, and seven of the short-answer questions correctly. Any questions answered incorrectly are worth zero points. How many points did Amber score on her history test?

1.

2. The rectangle below has a width of 15 and a height of 8. The rectangle is cut into four smaller rectangles by cuts along the dotted lines. What is the sum of the perimeters of the four smaller rectangles?

2.

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Problems 3 & 4

3. The product of the tens and units digit of a three-digit whole number is 12. The product of the tens and hundreds digits of the same three-digit whole number is 4. What is the greatest possible value of the three-digit whole number?

3.

4. Three busy beavers can build six dams in two days. If all busy beavers always work at the same constant rate, then how many days will it take two busy beavers to build eight dams?

4.

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Problems 5 & 6

5. Chauncey chooses a two-digit whole number. If Chauncey divides his number by two, the result will be a two-digit whole number, and if Chaunchey multiplies his number by two instead, the result will also be a two-digit whole number. How many different numbers could Chauncey have chosen?

5.

6. Mrs. Howard has six different prizes to award to students in her class this school week, which is every day from Monday through Friday. She wants to give at least one prize on each day of the school week. In how many different ways can Mrs. Howard select which prizes to award on each day of the school week?

6.

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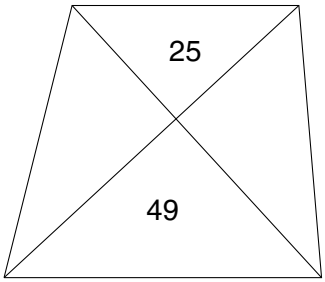
Problems 7 & 8

7. Paul and his daughter Leah share the same birthday. When Leah turned one year old, Paul turned thirty-six years old, so Paul's age was a multiple of Leah's age. If Paul and Leah both live to be at least 100 years old, on how many other birthdays will Paul's age be a multiple of Leah's age?

7.

8. As shown below, diagonals are drawn in a trapezoid to create four triangles. The two triangles that have sides which are also bases of the trapezoid have areas of 25 and 49. What is the area of the trapezoid?

8.

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# Team Round 12224

School or Team

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|---|---|---|---|---|
| 1. | 2. | 3. | 4. | 5. |
| 6. | 7. | 8. | 9. | 10. |

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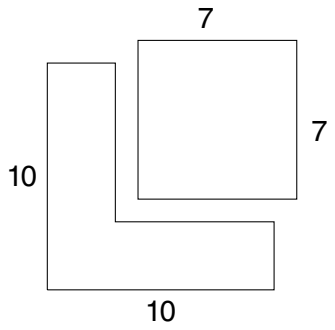
1. Bhairav is buying candy with coins from his piggy bank. Bhairav has a number of coins worth 25 cents each. He buys six gobstoppers for 10 cents each, nine lemondrops for 5 cents each, and ten chocolate-covered peanuts for 12 cents each. How many 25 cent coins does Bhairav spend on candy?

2. What is the sum of the first 40 positive whole numbers that are odd but are not multiples of 5?

3. Emma is giving out butterfly stickers to her friends. Emma gave Stella half of her stickers. Then Emma gave Allison one dozen stickers. Finally, Emma gave Denise half of the stickers she had left. After Emma gave butterfly stickers to her friends, she had seven butterfly stickers left. How many butterfly stickers did Emma have before she gave any to her friends?

4. From a square piece of paper that has sides measuring 10 inches, two cuts are made to remove a square of side length 7 inches, as shown below. What is the area, in square inches, of the L-shaped piece of paper that remains?




5. At the school fair, prize tokens are worth either 1 point, 2 points, or 4 points. Mr. Tolkein has ten prize tokens worth a total of 25 points. What is the greatest possible number of tokens worth 2 points that Mr. Tolkein could have?

6. Rafael chose a three-digit positive whole number. He then created a new number by reversing the digits of his original number, swapping the units and hundreds digit. For example, if Rafael chose the number 247, the new number would be 742. After Rafael created the new number, he multiplied the new number by the original number, and his product was 394695. What is the sum of Rafael's new number and old number?

7. In a list of positive whole numbers, each number after the first two numbers is the sum of the previous two numbers in the list. The sixth number in the list is 2022. What is the greatest possible value for the first number in the list?

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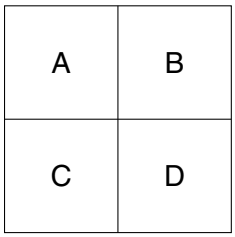
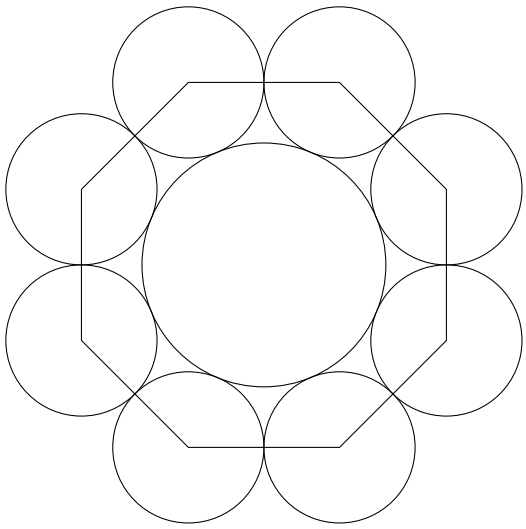
8. A square is divided into four smaller squares, as shown below. Colin wants to color the four smaller squares. He has a red marker, a blue marker, and a green marker. But he does not want any two of the smaller squares that share a side to be colored the same color. How many different ways can Colin color the smaller squares?




9. What is the sum of the tens digit and the units digit of $2022^{2022}$?

10. The regular octagon below has a side of length 40. Eight conguent small circles are drawn with centers that are also vertices of the octagon. Each small circle is tangent to two other small circles. Finally, a large circle is drawn that is tangent to each of eight smaller circles, as shown below. To the nearest whole number, what is the area of the large circle?



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# Answers 12224

## Sprint Round

1. C
2. A
3. D
4. D
5. A
6. A
7. E
8. D
9. B
10. E
11. B
12. E
13. A
14. E
15. D
16. B
17. D
18. A
19. D
20. C
21. B
22. D
23. E
24. D
25. C
26. D
27. C
28. A
29. E
30. C

## Target Round

1. 86
2. 92
3. 226
4. 4
5. 15
6. 1800
7. 3
8. 144

## Team Round

1. 9
2. 2000
3. 52
4. 51
5. 6
6. 1272
7. 669
8. 18
9. 12
10. 3270

## Number Sense

1. 243
2. 162
3. 1
4. 60
5. 170
6. 4
7. 480
8. 714
9. 12
10. $[2337, 2583]$
11. 682
12. 205
13. 1
14. 990
15. 361
16. 891
17. 1620
18. 347
19. 16
20. $[85072, 94026]$
21. 1150
22. $\frac{6}{11}$
23. 21
24. 961
25. 5
26. 17
27. 105
28. $\frac{1}{2}$
29. 1760
30. $[84635, 93543]$
31. 5049
32. 3025
33. 70
34. 56
35. 1209
36. 46
37. 19
38. 385
39. 29
40. $[487, 537]$
41. 4
42. 2464
43. 671
44. 178
45. 160
46. 14
47. 7575
48. 100
49. 800
50. $[113989, 125987]$
51. 13287
52. 135
53. 11
54. 710
55. 3996
56. 19
57. 26
58. 30
59. 7872
60. $[25918, 28645]$
61. 15
62. $\frac{2}{15}$
63. $37\frac{1}{36}$
64. 333
65. 52
66. 12375
67. 225
68. 11424
69. 1600
70. $[877345, 969697]$
71. 60
72. 34225
73. 125
74. 1728
75. 2197
76. 6
77. 81
78. 24990
79. 108
80. $[1595636, 1763596]$

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# Solutions 12224

## Sprint Round Solutions

1. The values of the products are $4 \times 8 = 32$, $5 \times 5 = 25$, $11 \times 3 = 33$, $2 \times 15 = 30$, and $7 \times 4 = 28$. Only 32 and 33 are greater than 31, for a total of $\boxed{2}$.

2. Hayden spent a total of $\$4+\$3+\$3+\$4+\$5 = \$19$ on snacks, so the number of dollars that remained was $\$20 - \$19 = \boxed{\$1}$.

3. There are 3 instances of $10 + 20 + 30 + 40 = 100$, for a total of $3 \times 100 = 300$. There are 4 instances of $1 + 5 + 9 = 15$, for a total of $4 \times 15 = 60$. Altogether, the sum is $300 + 60 = \boxed{360}$.

4. The area of the first rectangle is $8 \times 6 = 48$. The area of the second rectangle is $5 \times 9 = 45$. The total area of both rectangles is $48 + 45 = \boxed{93}$.

5. Since there are 60 minutes in 1 hour, a time 60 minutes after the start of the movie would be 8:18 PM, and a time $60 + 60 = 120$ minutes after the start of the movie would be 9:18 PM. An additional $124 - 120 = 4$ minutes would bring the time to $\boxed{\text{9:22 PM}}$.

6. Michelle has 5 gallons, which is $5 \times 4 = 20$ quarts, or $20 \times 2 = 40$ pints. If Michelle sells 40 pints for \$2 per pint, she will receive a total of $40 \times \$2 = \boxed{\$80}$.

7. If day 1 of January was a Saturday, then day 2 was a Sunday, day 3 was a Monday, day 4 was a Tuesday, and day 5 was a Wednesday. If day 5 was a Wednesday, then so are days $5 + 7 = 12$, $12 + 7 = 19$, and $19 + 7 = 26$. The last Wednesday occurs on day $\boxed{26}$.

8. All of the answer choices are less than 100, so it is only necessary to check for divisibility by primes less than $\sqrt{100} = 10$. These primes are 2, 3, 5, and 7. None of 73, 79, 83, and 97 are divisible by 2, 3, 5, or 7, but $91 = 7 \times 13$, so the number that is not prime is $\boxed{91}$.

9. Considering numbers that differ by 8, $10 \times 18 = 180$ is too large, but $2 \times 10 = 20$ is too small. While $5 \times 13 = 65$ is closer, although still too small, $6 \times 14 = 84$, and the sum of 6 and 14 is $\boxed{20}$.

10. Two of the triangles created can be placed together to form a square of side length $12 \div 2 = 6$. The area of one of these squares is $6 \times 6 = 36$, so the area of 4 triangles, which is 2 squares, is $2 \times 36 = 72$. The area of the original square is $12 \times 12 = 144$, and the area of the four triangles is 72, so the area of the smaller square is $144 - 72 = \boxed{72}$.

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11. The prime factorization of 105 is $3 \cdot 5 \cdot 7$. The prime factorization of 165 is $3 \cdot 5 \cdot 11$. The factors that divide 165 but do not divide 105 are those that are multiples of 11. That is $1 \cdot 11 = 11$, $3 \cdot 11 = 33$, $5 \cdot 11 = 55$, and $3 \cdot 5 \cdot 11 = 165$, for a total of $\boxed{4}$.

12. The number of acorns Colleen collected on the first 5 days is $3 + 7 + 11 + 15 + 19$. This is $5 \cdot 11 = 55$. Similarly, the number of acorns Colleen collected on the second 5 days, or days 6 through 10, is $23 + 27 + 31 + 35 + 39$, or $5 \cdot 31 = 155$. So in the first 10 days, she collected a total of $55 + 155 = 210$ acorns. On day 11, she collected 43 more, bringing her total to $210 + 43 = 253$, and on day 12 she collected 47 more, bringing her total to $253 + 47 = 300$. Her total collection exceeded 299 acorns on day $\boxed{12}$.

13. The area of a right triangle is half the product of the lengths of the legs, so the area of the original piece of paper is $\frac{1}{2} \cdot 40 \cdot 30 = 600$ square centimeters. The area of the piece of paper that was cut out and removed is $\frac{1}{2} \cdot 20 \cdot 15 = 150$ square centimeters. The area of the remaining piece of paper, in square centimeters, is $600 - 150 = \boxed{450}$.

14. The fee of \$5 covers pays for 1 of the 17 miles, so $17 - 1 = 16$ miles cost \$1.25 per mile, for an additional $16 \times \$1.25 = \$20$. Adding in the tip, the total cost of the Ryde ride is $\$5 + \$20 + \$4 = \boxed{\$29.00}$.

15. The number 9! can be written as $9 \cdot 8 \cdot 7!$, and the number 8! can be written as $8 \cdot 7!$. Then the expression becomes $(9 \cdot 8 \cdot 7! + 8 \cdot 7! + 7!) \div (7!)$, or $7! \cdot (9 \cdot 8 + 8 + 1) \div (7!)$. The 7! in the dividend and divisor cancel, leaving $9 \cdot 8 + 8 + 1 = \boxed{81}$.

16. If the perimeter for both rectangles is the same, then the width of the rectangle with height 7 exceeds the width of the rectangle of height 10 by $10 - 7 = 3$. Let the width of the rectangle with height 10 be $W$, and the rectangle with height 7 must have width $W + 3$. If the area of the rectangle with height 7 is greater, then $7(W + 3) - 10W = 27$, or $21 - 3W = 27$, which is impossible, since $W$ must be positive. Therefore, the area of the rectangle with height 10 is greater, so $10W - 7(W + 3) = 27$, or $3W - 21 = 27$. Then $W = 16$, and the perimeter of one of the rectangles is $2 \cdot (10 + 16) = \boxed{52}$.

17. The remaining four numbers are 5, 7, 8, and 9. If the two additional numbers in the middle column are both odd, then the sum of the numbers in the middle column will be even, which is not permitted. Similarly, if the two additional numbers in the bottom row are both odd, then the sum of the numbers in the bottom row will be even, which is not permitted. Thus the only remaining even number, 8, must be placed into the middle column and bottom row.

    The remaining 3 odd numbers may be placed into the remaining three squares in any way. There are 3 possibilities for the square in the middle column of the middle row. Once that number is selected, there are 2 possibilities for the rightmost square in the middle row, and then there is only 1 possibility for the rightmost square square in the bottom row. Therefore the total number of ways to place the remaining four numbers is $3 \times 2 \times 1 = \boxed{6}$.

18. The number $17\sqrt{7}$ is equal to $\sqrt{17^2} \cdot \sqrt{7}$. This is $\sqrt{289} \cdot \sqrt{7}$, or $\sqrt{289 \cdot 7} = \sqrt{2023}$. Because $45^2 = 2025$ and $44^2 = 1936$, the number $\sqrt{2023}$ is nearest in value to $\boxed{45}$.

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19. An increase of 20% multiplies the value of the stock by $1 + 0.20 = 1.2$. Similarly, an increase of 25% multiplies the value of the stock by $1 + 0.25$, and an increase of 10% multiplies the value of the stock by $1 + 0.1$. But a loss of 40% multiplies the value of the stock by $1 - 0.4$. In total, the value of the stock since Stan made his purchase was multiplied by $(1 + 0.25)(1 + 0.2)(1 + 0.1)(1 - 0.4)$. This is $1.25 \cdot 1.2 \cdot 1.1 \cdot 0.6$, or $1.5 \cdot 0.66$, which is 0.99. This is equivalent to multiplying the value by $1 - 0.01$, which is a decrease of $\boxed{1\%}$.

20. This could be evaluated directly, but $a^3+3a^2b+3ab^2+b^3$ is equal to $(a+b)^3$. Therefore, $9\clubsuit5 = (9+5)^3$, and $14^3 = 14 \times 196$, or $14 \cdot 200 - 14 \cdot 4 = \boxed{2744}$.

21. If $S$ is the first letter, then there are $4! = 24$ ways to arrange *NOWS* in the remaining four positions. Similarly, if $S$ is the last letter, then there are $4! = 24$ ways to arrange *SNOW* in the remaining four positions. Summing these results yields $24 + 24 = 48$ possible arrangement. But this double-counts arrangements that both begin an end with the letter $S$. If $S$ is both the first and last letter, there are $3! = 6$ ways to arrange *NOW* in the remaining three positions. These 6 arrangements have been counted twice, so the total number of possible arrangements is $48 - 6 = \boxed{42}$.

22. When Isaac passes the sign, it is 12 minutes after noon, or 12:12 PM. At that time, Henry has ridden his bike for $\frac{12}{60} = \frac{1}{5}$ of an hour after he passed the sign, and is $\frac{1}{5} \cdot 10 = 2$ miles ahead of Isaac. Since Isaac rides his bike at 12 miles per hour and Henry rides his bike at 10 miles per hour, Isaac travels $12 - 10 = 2$ miles more than Henry in one hour, and he will catch up to Henry exactly one hour after he passes the road sign, which is $\boxed{\text{1:12 PM}}$.

23. Because 35 is a multiple of 5, the product is a multiple of 5. So, if $35 \cdot 36 \cdot 37 \cdot 38 \cdot 39 = 25k + r$ for some quotient $k$ and some remainder $r$, then, dividing both sides by 5, $7 \cdot 36 \cdot 37 \cdot 38 \cdot 39 = 5k + \frac{r}{5}$. The remainder of $7 \cdot 36 \cdot 37 \cdot 38 \cdot 39$ divided by 5 is determined by the units digit of the product, which is the units digit of $7 \cdot 6 \cdot 7 \cdot 8 \cdot 9$, or 8. Therefore, $\frac{r}{5}$ is 3, so $r$, the remainder when the product is divided by 25, is $3 \cdot 5 = \boxed{15}$.

24. Taken together, the four curved corners of the frame form a circle of radius 7. Using $\frac{22}{7}$ as an approximation for $\pi$, the area of this circle is $7^2 \cdot \frac{22}{7} = 154$. The remaining portions of the frame are two rectangles of width 15 and height 7, which together have an area of $2 \cdot 7 \cdot 15 = 210$, and two rectangles of height 10 and width 7, which together have an area of $2 \cdot 10 \cdot 7 = 140$. To the nearest whole number, the area of the frame, in square centimeters, is $154 + 210 + 140 = \boxed{504}$.

25. After creating a set of three beads, regardless of color, the greatest number of unmatched beads is 2 beads of each of the other 2 colors, for a total of 4 unmatched beads. Therefore, to guarantee 6 groups of beads, Stella would need to remove the $6 \cdot 3 = 18$ beads for those 6 groups plus the maximum of 4 unmatched that would remain after creating the sixth group, for a total of $6 \times 3 + 4 = \boxed{22}$.

26. The number $2^{30}$ can be expressed as $\left(2^7\right)^4 \cdot 2^2$, or $128^4 \cdot 4$. Because $5^{12} = \left(5^3\right)^4$, or $125^4$, this number is greater than $5^{12}$. Now consider $5^{13}$, or $125^4 \cdot 5$, and the quotient when divided by $2^{30}$. This quotient is $\left(\frac{125}{128}\right)^4 \cdot \frac{5}{4}$. If this is greater than 1, then $5^{13} > 2^{30}$. The quotient is greater than 1 if $\left(\frac{125}{128}\right)^4 > 0.8$. Since $\frac{125}{128} > 0.95$ and $0.95^4 > 0.8$, $\left(\frac{125}{128}\right)^4 > 0.8$, and the smallest power of 5 that is greater than $2^{30}$ is $\boxed{5^{13}}$.

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27. To win the game, on his last turn Micah needs to leave exactly 1 jellybean in the bag. This can be guaranteed if, on the preceding turn for Noah, there are exactly 8 jellybeans in the bag. This is because no matter the number of jellybeans Noah draws from the bag, Micah can draw a number that, when combined with the number Noah drew, totals 7 jellybeans. If Noah draws 6, Micah draws 1. If Noah draws 5, Micah draws 2. And so on.

Exactly 8 jellybeans in the bag can be guaranteed if, one turn for Noah and Micah earlier, there are $8 + 7 = 15$ jellybeans in the bag. And as long as the number of jellybeans in the bag is 1 more than a multiple of 7 prior to Noah taking a turn, the number of jellybeans remaining in the bag can be reduced by Micah to a smaller number that is also 1 more than a multiple of 7. Thus, on his first turn Micah needs to reduce the number of jellybeans in the bag to a number that is 1 more than a multiple of 7. Since $7 \cdot 14 = 98$, Micah needs to reduce the number of jellybeans in the bag to $98 + 1 = 99$, so the number of jellybeans he should draw is $100 - 99 = \boxed{1}$.

28. Let the width of the rectangle be $W$ and the height of the rectangle be $H$. Then the area of the rectangle is $WH$. The two triangles that share a vertical angle where the two segments intersect are similar. The base of one triangle is $\frac{H}{2}$ and the base of the other is $H$. By similarity, the altitudes, which sum to $W$, are $\frac{W}{3}$ and $\frac{2W}{3}$. Therefore the area of the smaller of the two triangles is $\frac{1}{2} \cdot \frac{H}{2} \cdot \frac{W}{3} = \frac{HW}{12}$. The third triangle, opposite the quadrilateral, and the larger triangle must have areas that sum to $\frac{HW}{2}$, so the area of all three triangles is $\frac{HW}{2} + \frac{HW}{12} = \frac{7HW}{12}$. Thus the area of the quadrilateral is $HW - \frac{7HW}{12} = \frac{5HW}{12}$. Since $\frac{5HW}{12} = 45$, $HW = 108$. The area of the three triangles is $\frac{7}{12} \cdot 108 = 63$, and $6 + 3 = \boxed{9}$.

29. Suppose the four-digit number is $ABCD$. Because no number with four of the same digit has a product of the digits that is equal to the sum of the digits, at least one digit must be less than the greatest digit. Additionally, no digit can be 0, since that would create a product of 0, and the sum would be greater. Since the order of the digits does not affect the value of the sum or product, let $A \geq B \geq C \geq D$, with $A > D$. Then $A + A + A + A > A + B + C + D$, so $A + A + A + A > A \cdot B \cdot C \cdot D$. That is, $4A > A \cdot B \cdot C \cdot D$, or $4 > B \cdot C \cdot D$.

If $B \cdot C \cdot D = 1$, then $B$, $C$, and $D$ would all be 1, and $A + 1 + 1 + 1 = A \cdot 1 \cdot 1 \cdot 1$, which is impossible. If $B \cdot C \cdot D = 3$, then two of $B$, $C$, and $D$ would be 1 and the other would be 3, so $A + 1 + 1 + 3 = A \cdot 1 \cdot 1 \cdot 3$, or $A + 5 = 3A$. Then $A = \frac{5}{2}$, which is also impossible. Finally, if $B \cdot C \cdot D = 2$, then two of $B$, $C$, and $D$ would be 1 and the other would be 2, so $A + 1 + 1 + 2 = A \cdot 1 \cdot 1 \cdot 2$, or $A + 4 = 2A$. Then $A = 4$, which is possible. Therefore the only four-digit positive whole numbers that are possible are permutations of the digits 1, 1, 2, and 4. Since there are two instance of the digit 1, the number of distinct arrangements is $\frac{4!}{2!} = \boxed{12}$.

30. The digits to the right of the octal point in base 8 have a value of $\frac{1}{8}$, $\frac{1}{64}$, $\frac{1}{512}$, and so on, with each place being $\frac{1}{8}$ the value of the place immediately to the left. The non-repeating portion of the number has a value of $1 \cdot \frac{1}{8} = \frac{1}{8}$. The repeating portion that has places occupied by a 5 has the value $\frac{5}{64} + \frac{5}{64^2} + \frac{5}{64^3} + \cdots$. This is an infinite geometric series with a first term equal to $\frac{5}{64}$ and a ratio of $\frac{1}{64}$. The repeating portion that has places occupied by a 2 has the value $\frac{2}{512} + \frac{2}{512 \cdot 64} + \frac{2}{512 \cdot 64^2} + \cdots$. This is an infinite geometric series with a first term equal to $\frac{2}{512} = \frac{1}{256}$ and a ratio of $\frac{1}{64}$. The sum of both of these series is $\frac{\frac{1}{256} + \frac{5}{64}}{1 - \frac{1}{64}} = \frac{\frac{1}{256} + \frac{20}{256}}{\frac{252}{256}}$. This is $\frac{21}{252} = \frac{1}{12}$. Therefore the value is $\frac{1}{8} + \frac{1}{12} = \frac{5}{24}$, and $5 + 24 = \boxed{29}$.

## Target Round Solutions

1. Amber's score on the multiple-choice portion of the test was $17 \times 3 = 51$, and her score on the short-answer portion of the test was $7 \times 5 = 35$, so her score on the entire test was $51 + 35 = \boxed{86}$.

2. The perimeter of the original rectangle is $15 + 15 + 8 + 8 = 46$. All of these edges exist on the four smaller rectangles. When the long cut parallel to the side of length 15 is made, edges are created with a total length of $15 + 15 = 30$. When the short cut parallel to the side of length 8 is made, edges are created with a total length of $8 + 8 = 16$. Therefore, the total perimeter of the four rectangles is $46 + 30 + 16 = \boxed{92}$.

3. If the product of the tens and hundreds digits is 4, then the tens and hundreds digits are either 4 and 1 in some order, or 2 and 2. If the hundreds digit is 4, then the tens digit is 1, and the units digit would be $12 \div 1 = 12$, which is impossible. If the hundreds digit is 1, then the tens digit is 4, and the units digit would be $12 \div 4 = 3$, so the number would be 143. If the hundreds digit is 2, then the tens digit is 2, and the units digit is $12 \div 2 = 6$, so the number would be 226. Of the two possibilities, the greatest is $\boxed{226}$.

4. If 3 busy beavers can build 6 dams in 2 days, then 3 busy beavers building 8 dams would take $2 \cdot \frac{8}{6} = \frac{8}{3}$ days. However, because only 2 busy beavers are working, not 3, the work will take $\frac{3}{2}$ as long, so the number of days required is $\frac{8}{3} \cdot \frac{3}{2} = \boxed{4}$.

5. Because the quotient when the number is divided by 2 is a whole number, the number must be even. Additionally, since the quotient is a two-digit whole number, the quotient must be at least 10, so the number must be at least 20. Additionally, because the product when the number is multiplied by 2 is a two-digit whole number, the product must be no more than 98, so the number must be no more than 49. The number of even numbers between 20 and 49, including 20, is $\frac{48-20}{2} + 1 = \boxed{15}$.

6. There are 5 days in total and 6 different prizes, and every day must have at least 1 prize, so there must be exactly 1 day that has 2 prizes, and all other days will have 1 prize. There are 5 ways to select the day that has 2 prizes. Additionally, there are $\binom{6}{2} = 15$ sets of two prizes that may be awarded on that day. Finally, there are $4 \cdot 3 \cdot 2 \cdot 1 = 24$ ways to assign the remaining four prizes to the four remaining days. Therefore, the total number of ways to select which prizes to award each day is $5 \cdot 15 \cdot 24 = \boxed{1800}$.

7. Paul will always be $36 - 1 = 35$ years older than Leah, and if Paul's age is a multiple of Leah's age, then the difference between their ages, 35, must also be a multiple of Leah's age. Leah's age must be a divisor of 35, so Paul's age will be a multiple of Leah's age when Leah is 5, 7, and 35, and the number of times this occurs is $\boxed{3}$.

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8. Let the shorter base of the trapezoid be $a$ and the longer base of the trapezoid be $b$. Additionally, let the altitude from the shorter base to the intersection of the diagonals be $c$ and the altitude from the longer base to the intersection of the diagonals be $d$. Then $\frac{1}{2} \cdot ac = 25$, and $\frac{1}{2} \cdot bd = 49$, or $ac = 50$, and $bd = 98$. The area of the trapezoid is $\frac{1}{2} \cdot (a+b)(c+d)$, or $\frac{1}{2} \cdot (ac + ad + bc + bd)$. Since $bd$ and $ac$ are known, this is $\frac{1}{2}(148 + ad + bc)$. The triangles of area 49 and 25 are similar, so $\frac{a}{c} = \frac{b}{d}$, so $ad = bc$. Additionally, $adbc = 50 \cdot 98$, and since $ad$ and $bc$ are equal, each is $\sqrt{50 \cdot 98} = 70$. Thus the total area of the trapezoid is $\frac{1}{2}(148 + 70 + 70) = \boxed{144}$.

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## Team Round Solutions

1. The cost of the gobstoppers is $6 \times 10 = 60$ cents, the cost of the lemondrops is $9 \times 5 = 45$ cents, and the cost of the chocolate-covered peanutes is $10 \times 12 = 120$ cents each. The total cost of his candy purchases is $60 + 45 + 120 = 225$ cents. The number of 25 cent coins needed is $225 \div 25 = \boxed{9}$.

2. In each range of 10 units digits, the sum will include 4 the four numbers that have a units digit of 1, 3, 7, and 9. The value of $1 + 3 + 7 + 9$ is 20. The value of $11 + 13 + 17 + 19$ is 60. The value of $21+23+27+29$ is 100. This continues, and the final group of 4 is the 37th through 40th odd numbers, $91 + 93 + 97 + 99 = 380$. The total sum is $20 + 60 + 100 + 140 + 180 + 220 + 260 + 300 + 340 + 380$, which is $\boxed{2000}$.

3. Before Emma gave half her stickers to Denise, she had $7 \times 2 = 14$ stickers. Before she gave Allison one dozen stickers, she had $14 + 12 = 26$. Before she gave Stella half of her stickers, the number of stickers Emma had was $26 \times 2 = \boxed{52}$.

4. The area of the original piece of paper is $10 \times 10 = 100$ square inches. The area of the square that was removed is $7 \times 7 = 49$ square inches. Therefore, the area of the L-shaped piece of paper, in square inches, is $100 - 49 = \boxed{51}$.

5. First, note that since 25 is an odd number, Mr. Tolkein must have at least 1 token worth 1 point. Then we work backwards from 9 tokens worth 2 points. We find that 9 tokens worth 2 points does not work because $9 \cdot 2 + 1 = 19$. We also find that 8 tokens worth 2 points does not work because even with using a token worth 4 points, the total is only $8 \cdot 2 + 1 + 4 = 21$. Similarly, 7 tokens worth 2 points does not work as the max total can only be $7 \cdot 2 + 1 + 2 \cdot 4 = 23$. However, 6 tokens worth 2 points does work because Mr. Tolkein can also have 3 tokens worth 4 points and a token worth 1 point, where the total value sums to $6 \cdot 2 + 3 \cdot 4 + 1 = 25$. Therefore, the greatest possible number of tokens worth 2 points that Mr. Tolkien could have is $\boxed{6}$.

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6. We can approach the problem with clues from divisibility and estimation. Because 394695 is a multiple of 5 (but not 25), either the hundreds or units digit of Rafael's number (but not both) must be a 5. Suppose 5 is the hundreds digit. Then because the units digit of the product is a 5, the units digit of his original number must be odd (since an even number would give the product a units digit of 0) and not equal to 5.

   Next, we can note that a number with 5 as the hundreds digit is at least 500 but less than 600. Then we find that the other number should be at least 600 but less than 800. This narrows down the units digit to be 7. Finally, since 394695 is a multiple of 3, the original number must also be a multiple of 3. Because $5+7=12$, the tens digit must be a multiple of 3. Trying $507 \times 705$, $537 \times 735$, $567 \times 765$, and $597 \times 795$, the product of 537 and 735 is 394695, and $537+735=\boxed{1272}$.

7. Let the first number in the list be $a$ and the second number in the list be $b$. Then the third number is $a+b$, the fourth is $b+(a+b)=a+2b$, the fifth is $(a+b)+(a+2b)=2a+3b$, and the sixth is $(a+2b)+(2a+3b)=3a+5b$. Thus $3a+5b=2022$. To maximize $a$, $5b$ must be as small as possible. But $2022-5b$ must be a multiple of 3. Since 2022 is a multiple of 3, then $b=3$ is the least positive value of $b$ that makes $2022-5b$ a multiple of 3. So $2022-5\cdot 3=2007$ is $3a$, and the greatest possible value of $a$ is $\frac{2007}{3}=\boxed{669}$.

8. Because no two squares that share a side may be colored the same color, at least 2 of the 3 colors must be used. If 2 colors are used, one color must be used to color the squares labeled $A$ and $D$, and the other color must be used to color the squares labeled $B$ and $C$. There are 3 choices for the first color and 2 choices for the second, for a total of 6 possibilities when using 2 colors.

   If 3 colors are used, one color must be used to color either squares $A$ and $D$, or squares $B$ and $C$. In either case, each of the remaining 2 colors must be used for one of the two remaining squares. If squares $A$ and $D$ are colored the same color, there are 3 choices for that color, then 2 choices for the color of square $B$, and 1 choice for the color of square $C$, for a total of $3\cdot 2\cdot 1=6$ possibilities. Similarly, if squares $B$ and $C$ are colored the same color, there are also 6 possibilities. Altogether, the total number of possible colorings is $6+6+6=\boxed{18}$.

9. The only portion of the product that affects the tens digit and units digit of the result is the tens digit and units digit of each term of the product, so it is only necessary to find the tens and units digit of $22^{2022}$. Each power of 22 after the first is a multiple of 4. There are only 20 multiples of 4 that are not multiples 5, as no power of 22 can be a multiple of 5, so the sequence of tens and units digits of powers of 22 beyond $22^2$ can be at most 20 terms before a term repeats.

   Powers 1 through 5 have tens and units digits of $22, 84, 48, 56, 32$. Then $22^{10}$ has the same tens and units digits as $32\cdot 32=1024$, and $22^{20}$ has the same tens and units digits as $24\cdot 24=576$. Power 21 has a tens and units digit of 72, and for power 22 the tens and units digits are 84, which are identical to the tens and units digits for $22^2$. Therefore the tens and units digits repeat every 20 terms. Since $\frac{2022}{20}$ has a remainder of 2, the tens and units digits will be the same as those for $22^2$ and $22^{22}$, which are 84, and $8+4=\boxed{12}$.

10. Label the octagon $ABCDEFGH$, and let $r$ be the length of the radius of the smaller circles and $R$ be the length of the radius of the bigger circle. Because two radii of the smaller circles are equal to a side length of the octagon, then $r$ is equal to $\frac{40}{2} = 20$. Consider diagonal $AE$ of the octagon. This passes through the center of the large circle and two points where the large circle is tangent to one of the smaller circles. The length of $AE$ is equal to $r + 2R + r = 2R + 40$. $AE$ is also the hypotenuse of right triangle $AEF$. The length of $AF$ is equal to the sum of the side of the octagon, 40, plus the twice the length of a leg of an isosceles right triangle with hypotenuse $FG$. This leg length of the isosceles right triangle is $\frac{40}{\sqrt{2}} = 20\sqrt{2}$, so the length of $AF$ is $40 + 2 \cdot 20\sqrt{2} = 40 + 40\sqrt{2}$. By the Pythagorean Theorem, $AE$ has a length of $\sqrt{40^2 + (40 + 40\sqrt{2})^2} \approx 104.525$. Therefore $R$ is approximately $\frac{104.525 - 2 \cdot 20}{2} \approx 32.2625$. The area of the large circle is then $32.2625^2 \cdot \pi$, which to the nearest whole number is $\boxed{3270}$.

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